

T-740-SPMM Project C: Report

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Executive Summary

In Phase C of the project, the team developed a formal proposal for the *Ísland.is* extension. Building on the outcomes of the previous phase, this proposal marks a change in focus from research and feasibility assessment to structured planning. This phase had different goals:

- Definition of the goal, strategy, value and current situation of the project
- Stakeholder Analysis.
- Scope definition and risk Analysis.
- Identification of needed resources and ways of working.

In the beginning, the team clearly defined the goal of the project alongside their definition of done and how the success can be measured. Strategy and value of the team align with those of the stakeholders: implementing a patient-centered extension to increase safety and guarantee better treatments abroad, while considering the current situation and identifying the main issues that need to be addressed.

In the second phase, the team identified and analyzed the stakeholders to better understand their roles, priority, expectations, and communication channels. After this, a stakeholder registry was created to provide a clearer and compact view of this information.

Next, the team defined the scope to better understand what should and should not be included, ensuring that the stakeholders' needs were met and resources were not wasted. To help with this process, the team created a product backlog and set a definition of done (DoD). Following this, a risk analysis was performed to identify and classify vulnerabilities and threats that could occur during the development phase, as well as ways to mitigate them.

Finally, the team performed a preliminary assessment of the required resources, including manpower, equipment, technical resources, and costs. To conclude this phase, the team defined the approach that will be used to ensure the project is delivered on time and meets all stakeholder requests.

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Chapter 1: Standardized Project Proposal

Name of project	<i>Ísland.is</i> Medical Summary
Team	Group 6 with members Julie Bauer, Kristján Orri Ragnarsson, Leonardo Salpini, Stella Stefánsdóttir and Tinna María Þorleifsdóttir.
Goal	The overarching goal of the project is better medical treatment outside Iceland so that travellers feel secure, relieved, and in control. The conditions of satisfactions will be reached when our goal is reached. Our definition of done is when Icelandic travellers are able to access and use their medical summary, and healthcare professionals can reliably access, interpret, and accept the medical information. Success will be measured by user testing, structured evaluation, surveys and interviews to ensure usability, security and clarity. After the project is done, Icelandic citizens and residents will be able to get better care abroad and gain confidence in receiving high-quality medical care abroad, while reducing stress and uncertainty in emergencies.
Strategy & Value	
Strategy Alignment	The project aims to align with Iceland digital integration strategy by developing patient-centered extension that integrates seamlessly with <i>Ísland.is</i> . The final extension will reduce cost, save time for healthcare professionals, reduce uncertainty for foreign healthcare professionals, and ensure accessible medical documentation.
Value Proposition	We propose several values: • Reduce cost to users from services related to obtaining medical information. • Save valuable time for Icelandic healthcare professionals from documentary visits. • Reduce time and uncertainty for foreign healthcare professionals to understand Icelandic medical records. • Provide Icelanders with comprehensive care by having accessible and reliable health documentation for use abroad, thus enhancing patient safety and ensuring continuity of care abroad. • Reducing probability of risks, saving time and support patient well-being by increasing patient centered care.
Current Situation	Currently, there is no standardized, secure way to carry medical summaries digitally abroad. Individuals often rely on printed documents, doctor's notes or MedicAlert IDs, if prepared in advance. This highlights a gap between having access to health records and receiving appropriate care when seeking medical attention abroad.
Project Governance	The owner of the project is Digital Iceland, operated by the Ministry of finance and economic affairs.

Stakeholder Analysis	The project's stakeholders are <i>Ísland.is</i> , interviewees and beta testers, development team of <i>Ísland.is</i> , project managers, external contractors, data protection authority, security, and Icelandic health authorities. For a detailed stakeholder analysis, see appendix A, for a stakeholder registry, see appendix B
Communication Strategy	The communication strategy is as follows: • Weekly email updates for progress tracking, risk flagging and emerging issues for invested stakeholders. • Monthly stakeholder meetings for feedback, milestone alignment and vision setting with respected stakeholders involved. • Frequent team and project meetings for regular assessments and coordinating sessions. • Internal security meetings to go over reviews with system security and compliance checks in accordance with the timeline. • Use of collaborative platforms such as Teams/Slack for real-time updates, file, sharing, discussion and Question&Answer sessions. • Face-to-face sessions for direct engagement between organization and project manager, project manager and stakeholders, and direct user and developers. • Surveys and scheduled interviews for collecting systematic feedback and information from testers and users to foster trust and value with the testers. • Reports and official letters for formal updates for regulators, compliance authorities and decision makers for formal validation and synergy with health authorities. • Official contracting channels for structured communication with external contractors and vendors engaged for specific phases rather than the entire project. This communication plan ensures transparency, collaboration, and engaged, informed and aligned stakeholders.
Scope	
Functionality	Functionalities that are in the scope of our project include: • Real-time/on the fly translation. • Presenting medical information in multiple formats, like pdf's and qr codes. • Secure authentication for access of information medical information. It is not in the scope of this project to present a full or extensive medical history records beyond essential summaries. See appendix C for Product Backlog of this project
Time	It is estimated that it takes about 1,5 to 2 years to completely develop this project. This includes design, implementation, testing, and regulatory compliance. Once completed, users can utilize the feature as an extension in <i>Ísland.is</i> .
Risk Analysis	
Assets	Assets include: • Medical data of users (personal medical summaries). • Users trust (confidence in usability and security). • System infrastructure (server, APIs). • Compliance (General Data Protection Regulation, Icelandic regulations). • Project resources (budget, timeline, expert personnel).

Vulnerabilities	Vulnerabilities are: • Medical health data downloaded without strong encryption. • Lack of user understanding; misuse of sharing data. • Lagging of infrastructure system or system being down. • Not meeting the requirements of standard regulations. • Delays in development, lack of resources or timeline risks.
Threats	Threats to our project are: • Cyberattacks; Hacking, ransomware, phishing. • Data breaches: Leaks of insider misuse. • Regulatory fines: Failure to comply with regulations or national laws. • System downtime: Technical failures during travel emergencies, and failure to prepare beforehand. • Negative publicity: Loss of trust from users due to security or usability issues. • Project risks: Staffing shortages, missed deadlines, cost overruns.
Risks	The risk assessment identifies potential challenges across technical, financial, organizational, social, project management, and external categories. Each risk was evaluated according to its probability and impact, and scored using a qualitative risk matrix. Risks with higher scores require immediate mitigation, while lower-scoring risks are monitored or accepted. See appendix D for a detailed risk analysis.
Mitigation	The mitigation outlines whether a risk must be actively addressed (Y=Yes), kept under observation (M=Monitor), or can be accepted with no further action (N=No). This structured approach was based on the Risk Matrix, see appendix D.2. This approach ensures that risks are managed proportionately. See appendix D for risk mitigation.
Resources	
Human Resources	The project requires several human resources: • Medical expertise to ensure accurate and clinically relevant summaries. • UX design expertise to ensure accessibility and user-friendliness. • Technical expertise in secure software development, data architecture, and interoperability. • Knowledge of the law and regulations to ensure compliance with laws.
Estimated Manpower	Estimation of five active personnel to work 15–20 months cohesively or 75–100 man-months.
Equipment	No new equipment is needed but there might be a need for new software (see next box for a description).
Technical	Technical resources that are required are: • A translation model. • Connection to medical databases. • Encryption/security.

Cost	The cost of the project is estimated as follows: • Employee payroll: 108–144 M ISK. • Compliance inspection: 4 M ISK. • 120 Hours of outside consulting for specialized external expertise: 4.8 M ISK. • Translation would be an ongoing cost (e.g. Azure S1 is \$2055 per month). This was estimated from various data-points Employee payroll from industry standard for a high performing programmer with some experience, compliance cost is based on list price for a compliance process times two so expecting the project to need to be compliant with two standards. The outside consulting is based on expert contractor pricing of 40000 ISK. per hour that is not unusual in Iceland for experts. and translation cost is taken straight from Azures price guide.
Ways of Working	
Approach	Our approach begins with defining what a medical summary should look like, using existing examples as reference. Once this structure is clear, we will arrange data access and develop methods to extract the relevant information into summaries. In parallel, we will consult experts to ensure ethical design, regulatory compliance, and user-centred development. Their insights will be incorporated throughout the process. The extension itself will be built step by step, starting with the main framework to guarantee that medical summaries can be securely displayed. This iterative process allows us to adjust early if needed. Key priorities include security and accurate English translation. Translations into less common languages, support for different display formats, and further extensions such as insurance alignment or alternative access methods (for example, family access) will be considered at later stages, depending on time and resources.
Method of Software Development	We recommend using an Agile approach, specifically the Scrum methodology, to ensure flexibility, transparency, and active stakeholder involvement. The project will be developed in short sprints, allowing for incremental improvements, continuous feedback, and the early detection of risks. By streamlining activities and prioritizing the most valuable features first, we can deliver meaningful results early on while keeping the project adaptable to new insights and requirements.

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Appendix A: Stakeholder Analysis

Parameter	Description
Stakeholder	<i>Ísland.is</i>
Internal or external	Internal
Organization and location	<i>Ísland.is</i> in Iceland
Role and responsibility in project	<i>Ísland.is</i> is the most important stakeholder, asking us to develop the system. They came up with the original idea and are responsible for guiding the development of the system. They are also responsible for answering questions that come up during the development, ensures alignment with national goal
Power	High – ultimate decision-making authority
Interest	High interest in the outcome.
Attitude	A neutral (3) attitude is expected. The attitude can change based on the progress of the project.
Priority	It is very important to keep them informed because they are our client.
Requirements / Expectations	We expect this stakeholder to provide continuous feedback and communicate smoothly. The stakeholder expects us to do a comprehensive user group analysis and UX goals, testing, interviews and feedback integration, meet up with them, project planning, hiring etc.; all necessary steps required to finish the project successfully.
Communication channel	We provide smaller updates through email. We hold regular monthly meetings where we ask questions that have come up, give them updates on our progress and get feedback on our progress.
Other data	Strategic partner; project depend on their satisfaction on deliverables

Parameter	Description
Stakeholder	Interviewees and bèta testers
Internal or external	External
Organization and location	Individual users/ test groups
Role and responsibility in project	Interviewees and bèta testers function as users of the system. Their responsibility is to give truthful and detailed feedback on usability, features and overall design
Power	Medium power; they do influence decision-making through their feedback in testing the system, but they do not have the ability to make final or overarching decisions
Interest	High interest, users benefit directly from usability and functionality
Attitude	Depends on the person; some might be negative (0) and some will be positive (1). Both are equally important and will be considered in the development process.
Priority	Not important to keep them informed; mainly engaged during testing and feedback cycles
Requirements / Expectations	The stakeholders expect intuitive and easy-to-use design, useful features, data privacy, system security and accurate and concise summaries
Communication channel	Email will be used to communicate with these stakeholders to schedule interviews and testing days, feedback surveys
Other data	Insights help validate user needs and guide design refinements

Parameter	Description
Stakeholder	Development team of <i>Ísland.is</i>
Internal or external	Internal
Organization and location	Digital Iceland
Role and responsibility in project	Build and implement the system; execute technical work
Power	Low in strategic decision making but high in implementation
Interest	High interest – responsible for technical success
Attitude	3 Positive – motivated to deliver
Priority	High - critical to be kept informed in every step. Influential stakeholders in delivering the project.
Requirements / Expectations	Deliver tasks on time and within scope, maintain strong internal communication and collaborate smoothly with other stakeholders
Communication channel	Daily meetings, emails, internal collaboration tools, face-to-face
Other data	Backbone of the project, success relies on their competence and coordination

Parameter	Description
Stakeholder	Project Managers
Internal or external	Internal
Organization and location	Part of the project team at Reykjavik University
Role and responsibility in project	Plan, organize, manage resources, monitor budget and ensure timely delivery
Power	High – oversee project execution
Interest	High accountable for project outcome
Attitude	4; We will streamline the project in a positive, yet critical manner
Priority	High importance, stakeholders need to be informed every step for need constant updates to adjust plans and resources
Requirements / Expectations	Sufficient resources to complete the process, support from the developer team, timelys updates on progress and risks, smooth coordination between stakeholders
Communication channel	Frequent meetings, emails, reports, and face-to-face check-ins
Other data	Act as a bridge between client, developers, and other stakeholders

Parameter	Description
Stakeholder	External Contractors
Internal or external	External
Organization and location	External Software development company/division under contract
Role and responsibility in project	Creating, updating and managing APIs and services that are contracted out and would need to be changed for the project.
Power	Low – only contribute technical support
Interest	Low – project not core to their mission
Attitude	2
Priority	Important only for necessary API changes
Requirements / Expectations	Clear contracts and deliverables, timely payment
Communication channel	Through the official channels used by the contracting agency
Other data	Provide specialised expertise but limited strategic involvement

Parameter	Description
Stakeholder	Data protection authority
Internal or external	external
Organization and location	Data protection authority of Iceland
Role and responsibility in project	Ensure the extension provides data privacy protection
Power	Low
Interest	Medium
Attitude	3
Priority	Medium priority
Requirements / Expectations	Compliance with GDPR and Icelandic privacy laws, data minimization, anonymization where possible, secure data handling practices
Communication channel	Official letters, report and email communication through formal channels
Other data	Regulatory body; involvement mainly during compliance checks, audits, or in case of reported data breaches

Parameter	Description
Stakeholder	Security
Internal or external	Internal
Organization and location	Digital Iceland
Role and responsibility in project	To make sure all information in the system is secure
Power	Medium, they do influence decision-making through their feedback, but they do not have the ability to make final or overarching decisions.
Interest	Medium; system security is central
Attitude	3
Priority	High, key in preventing vulnerabilities and ensuring trust in the system
Requirements / Expectations	Implementation of encryption, secure access management, regular penetration testing, compliance with national cybersecurity standards
Communication channel	Internal meetings, email updates, incident reporting systems
Other data	Works closely with developers and data protection authority to align privacy and security measures.

Parameter	Description
Stakeholder	Icelandic health authorities
Internal or external	External
Organization and location	Directorate of health, Primary health care in the capital, University hospital of Iceland, Ministry of Health
Role and responsibility in project	Provide standards, medical expertise, and approvals for compliance with health regulations
Power	Medium power in decision making, opinions influence final approval
Interest	High – outcome directly impacts citizens and healthcare services
Attitude	3
Priority	Medium informed during final deliverables and standard compliance
Requirements / Expectations	System must meet medical and regulatory standards, effective communication and approval processes
Communication channel	Email, official health authority/external communication system
Other data	Ensure the system/display is accepted in healthcare environments and trusted by professionals

Appendix B: Stakeholder Registry

Stakeholder	Role /responsibilities	Organization & location	Requirements /Expectations	Communication channel	Other data
<i>Ísland.is</i>	Client; came up with idea, guides development, answers questions	<i>Ísland.is</i> in Iceland	User analyses, testing, UX goals, meetings, planning, hiring	Email for updates, monthly meetings for feedback	Strategic partner; project depend on their satisfaction on deliverables
Interviewees & Bèta testers	System users providing detailed usability feedback	Individuals/test groups	Intuitive design, useful features, privacy, security, concise summaries	Email invitations, surveys, scheduled interviews	Insights help validate user needs and guide design refinements
Development team	Technical implementation and delivery of the system	Digital Iceland	Deliver on deadlines, communicate & cooperate well with other stakeholders	Meetings, emails, collaboration tools, face-to-face	Backbone of the project, success relies on their competence and coordination
Project managers	Plan, organize, manage resources & monitor budget	Reykjavík University	Sufficient resources, developer support, updates on progress/issues, risk assessor.	Frequent meetings, daily assessments, email, face-to-face	Act as a bridge between client, developers, and other stakeholders
External contractors	Manage APIs & external services under contract	External software development company/division	Payment	Official contracting channels	Provide specialized expertise but limited strategic involvement
Data protection Authority	Ensure GDPR and national privacy law compliance	Icelandic DPA	Ensure compliance with data protection standards	Official letters, reports, email	Regulatory body; involvement mainly during compliance checks, audits, or in case of reported data breaches

Stakeholder	Role/ responsibilities	Organization & location	Requirements/ Expectations	Communication channel	Other data
Security	Ensure system information security	Digital Iceland	Maintain system security	Internal security meetings, reports and emails	Works closely with developers and data protection authority to align privacy and security measures.
Icelandic Health Authorities	Review standards, medical expertise	Directorate of Health, Primary Health Care, Landspítali University Hospital of Iceland, Ministry of Health	Approve if meeting medical standards	Email, official health authority/ external agency channels.	Ensure the system/display is accepted in healthcare environments and trusted by professionals

Appendix C: Product Backlog

Name: Short name to describe the user story.

Description: User story; As a <user>, I want to <do something>, so that I can <some reason>.

Story Points: S = < 1 week of development, M = 1-2 weeks of development, L = 3-4 weeks of development, XL = > 4 weeks of development.

Acceptance Criteria: Conditions that must be met for the user story to be considered done.

ID	Name	Description	Story Points	Acceptance Criteria	Priority
1	Log in	As an Icelandic citizen I want to log in to <i>Ísland.is</i> so that I can see my medical summary.	L	User can enter valid <i>Ísland.is</i> credentials and granted access. After successful login, the user is redirected to their medical summary page. If the user enters invalid credentials, access is denied, and an error message is shown.	A
2	Choosing information	As an Icelandic traveler I want to choose the medical information that is on my summary so that I can show relevant information.	M	User can select/deselect items to have on summary. Vital items show a warning if deselected. On Save, the summary is displayed. If saving fails, the user sees a clear error and previous setting remain unchanged.	A
3	Warning	As an Icelandic traveler, I want the system to warn me before I remove any vital medical information from my summary, so that I don't accidentally delete critical details that a healthcare professional might need.	S	When the user selects to remove vital information, the system displays a warning message. The user must confirm or cancel the removal. If the user confirms, the info is removed, and summary is updated. If the user cancels, the info remains in the summary unchanged. If the info is non vital, the system removes it without showing a warning.	A
4	Changing language	As an Icelandic traveler I want to change the language of my medical summary so that I can travel with it abroad.	L	User can choose a language from a list of supported languages. Once selected, the medical summary is displayed immediately in that language. If translation fails, the user sees a clear error message and the last available version remains visible.	A

ID	Name	Description	Story Points	Acceptance Criteria	Priority
5	Download summary	As an Icelandic traveler, I want to download my medical summary so that I can travel with it and look it up offline.	S	User can download their summary. After downloading, the summary is stored securely on the device and accessible without internet. The app displays a confirmation message when successful. The summary is readable offline. If download fails, the user sees a clear error message and can retry.	A
6	Travel with summary	As an Icelandic traveler, I want to travel with a medical summary so that if something might happen, I have an up-to-date summary for the health-care professionals.	S	User can download their medical summary from <i>Ísland.is</i> . The summary includes up-to-date information. The summary is available offline once downloaded. User receives a confirmation message after successful download.	A
7	Simple and easy	As an Icelandic traveler with a low digital literacy, I want simple and easy steps in downloading it so that I can do it on my own or with minimal assistance.	S	The downloading option is clearly visible. The download process takes no more than 2-3 steps. Each step includes clear instructions with simple wording and/or icons. The system provides a confirmation message once the download is complete. If something goes wrong, the user receives a clear error message with guidance.	A
8	Changing language on the fly	As an Icelandic traveler, I want to be able to change the language of my medical summary on the fly so that doctors abroad can understand it even if it is not originally in their language.	L	User can select a different language for their summary from a list of supported languages. When selected, the summary is displayed in the chosen language immediately. If translation fails, the user sees an error message and the last successfully displayed version remains visible.	A
9	Real time update	As an Icelandic traveler, I want my medical information to update in real time whenever a doctor adds new information so that I can always have the medical summary up to date.	L	When a doctor adds/changes information, the user's medical summary in <i>Ísland.is</i> is updated within 1-2 minutes. The user gets a notification when new information has been added. The user can manually refresh their summary to check for new updates.	A
10	Fast translation	As an Icelandic traveler, I want the translation to be fast so that I can quickly show doctors my summary.	L	When the user requests a translation, the summary is displayed in the selected language within 2 seconds. If translation fails, the user sees a clear error message and can retry.	A

ID	Name	Description	Story Points	Acceptance Criteria	Priority
11	Certified information	As an Icelandic traveler, I want to ensure my healthcare information is certified so that the foreign healthcare professional can easily verify its authenticity.	M	The medical summary includes a digital certificate or seal issued by <i>Ísland.is</i> . Foreign doctors can see the certification mark on the document. The certification can be verified electronically. If the certification is invalid, the system clearly shows that.	A
12	Remove form summary	As an Icelandic traveler, I want to remove information on my medical information so that I can have what I want on the summary.	S	User can select and remove an item on the summary. The system updates the summary to no longer show the removed item. If the removed information is considered critical, the system displays a warning before final removal. User can confirm or cancel the removal after seeing the warning.	A
13	Two factor authentication	As an Icelandic traveler, I want two factor authentication when accessing sensitive data, so my information is protected.	M	User is prompted for a second factor after entering their password. Access to sensitive data is granted only if the second factor is correct. If the second factor is incorrect, the user cannot see the sensitive data and gets an error message. If the user does not complete the second factor, access is denied.	A
14	Log out	As an Icelandic traveler, I want to log out easily so that no one else can use my device to view my data.	S	User can log out by a “Log out” button. User gets redirected to the log in view.	A
15	Allergies	As an Icelandic traveler, I want my medical summary to include allergies so that I can inform foreign doctors quickly.	S	The allergies section is shown at the top of the summary when at least one allergy exists. Each allergy displays substance, reaction, severity and status. If no allergies are on record, the summary shows “No known allergies”.	A
16	Current prescription	As an Icelandic traveler, I want my medical summary to include current prescriptions so that doctors abroad know what I take.	S	Current prescriptions is shown clearly in the summary, with name, dose and frequency. If no prescriptions are on record, the summary shows “No prescriptions.”	A
17	Chronic conditions	As an Icelandic traveler, I want my medical summary to include chronic conditions so that I doctors abroad know about them.	S	Chronic conditions are shown clearly in the summary. If no chronic conditions are on record, the summary shows “No chronic conditions.”	A

ID	Name	Description	Story Points	Acceptance Criteria	Priority
18	Brood type	As an Icelandic traveler, I want my blood type on the medical summary so that doctors know it in emergencies.	S	Brood type is shown clearly in the summary.	A
19	Vaccination history	As an Icelandic traveler, I want vaccination history included so that doctors abroad know my status.	S	Vaccination section is shown in summary, with vaccine name, dose and date.	A
20	Emergency contact	As an Icelandic traveler, I want my emergency contacts list on y medical summary so that hospitals can reach them if needed.	S	An emergency contacts section is shown in summary, otherwise display “No emergency contacts on file”. Shows name, relationship, number and email.	A
21	Primary doctor information	As an Icelandic traveler, I want my primary doctor’s contact info on the medical summary so that foreign doctors can contact them.	S	A primary doctor section is shown, otherwise “No primary doctor on file”. Name, role, organization, phone, and email.	A
22	Encrypted	As an Icelandic traveler, I want my offline downloaded file encrypted so that no one can read it if my phone is lost.	XL	The downloaded offline summary is stored encrypted. Opening the summary requires user authentication	A
23	English	As an Icelandic traveler, I want to view my medical summary in English so that I can share it abroad.	M	User can select English from the language selector. Upon selection, the entire summary updates immediately to English. If translation fails, the app shows a clear error and keeps the last successfully displayed version.	A
24	Default language	As an Icelandic traveler, I want to select my preferred default translation language so that I don’t need to choose each time.	M	User can select a default translation language form a list of languages. When the user opens their medical summary, it automatically displays in the saved default language. User can temporarily switch languages in summary view, this does not change the default unless they save as default.	B
25	Preview	As an Icelandic traveler, I want to preview the original and translated version so that I can confirm accuracy.	L	User can view original and translated side by side. If section can’t be translated, the translated view shows “Not available – showing original.”	B
26	Read translated summary	As a doctor abroad, I want to see the translated medical summary so that I can treat the patient properly.	S	Doctor can open a translated summary.	A

ID	Name	Description	Story Points	Acceptance Criteria	Priority
27	QR code	As an Icelandic traveler, I want to generate a QR code of my summary so that doctors can scan it easily.	L	User can generate a QR code for the current summary with one action. The summary is shown in current selected language.	B
28	PDF	As an Icelandic traveler, I want to share my medical summary as a PDF so that I can email it if needed.	L	User can export to PDF. PDF uses the current language. User can open the PDF offline. User can share via email or save locally. If export fails, shows a clear error message and retry option.	A
29	Revoke shared files	As an Icelandic traveler, I want to revoke shared files after use so that my data isn't reused without my permission.	L	User can view a list of active shares with created time, expiry and last accessed. User can revoke any share and gets a success confirmation. Revoked links/QRs become immediately invalid. User can set an auto expire date.	B
30	Notify when out-dated	As an Icelandic traveler, I want to be notified when my medical summary is outdated so that I can update it.	M	User sees a date labeled "Last updated". User can refresh and the date changes.	B
31	Multiple summaries	As an Icelandic traveler, I want to store multiple summaries so that I can travel with my children's data.	XL	User can add multiple profiles and see them in a profile switcher. Adding a child requires guardian verification. Each profile stores its own selected sections, language and offline copy.	A
32	Fingerprint and face ID	As an Icelandic traveler, I want to be able to open my summary with fingerprint or face ID so that it's quick and secure.	L	User can open summary with face ID and fingerprint. After 3 failed attempts, they need to use PIN	A
33	PIN for offline version	As an Icelandic traveler, I want my offline summary to be encrypted with my device PIN so that only I can open it.	L	Offline summary is encrypted at rest. Opening the summary requires the device PIN. After 5 failed PIN attempts, access is temporarily locked for 5 minutes.	B
34	Usage history	As an Icelandic traveler, I want to see when and where my summary was last accessed so that I can detect misuse.	L	User can open Access history showing date and time. Offline views are logged locally and sync to history when online.	B
35	Delete offline version	As an Icelandic traveler, I want the option to delete my offline copy so that no traces remain on my device.	S	User can delete offline copy. On success, user sees "Offline copy deleted". If deletion fails, app shows clear error and user can retry.	A

ID	Name	Description	Story Points	Acceptance Criteria	Priority
36	Reminder to update	As an Icelandic traveler, I want reminders to update my summary before traveling so that I don't forget.	M	User can set a trip with start date and enable Update reminder. App schedules a reminder 5 days before the trip.	C
37	Notified when new data	As an Icelandic traveler, I want to be notified when new critical health data is available so that my summary stays relevant.	L	App detects critical changes. User receives a notification, showing what changed. User can refresh and the summary will update.	A
38	Confirmation after exporting	As an Icelandic traveler, I want confirmation after exporting/sharing my summary so that I know it was successful.	S	After exporting PDF, app shows "Exported" with file name, size and save location. Failures show clear error with Retry option.	A
39	Synced with my pages	As an Icelandic traveler, I want my summary synced with My Pages on Ísland.is so that I don't need to re-enter data.	L	Data is pulled automatically from My pages in Ísland.is. First open fetches latest data and shows Last updated, time and date.	A
40	Simple look	As a healthcare professional accessing the medical summary for the first time, I want the entire look to be intuitive so I that I know what I need to do when seeing it.	M	Critical info is clear and on top, allergies, medication, conditions, blood type, vaccinations. Clear section headings with icons.	A
41	Only important information	As a healthcare professional looking at the medical summary, I want important information only so that I don't waste too much time looking at it.	S	Summary shows critical info first; allergies, current medication, conditions, blood type, vaccinations, emergency contacts. Each section is clearly labeled.	A
42	Information on one page	As a healthcare professional providing emergent care, I want the data to load in one page/ one screen format and uncluttered setting so that I can read the information comprehensively.	S	Summary. opens to a single screen view, with critical info; allergies, current medication, conditions, blood type, vaccinations, emergency contacts. Layout fits without scrolling.	A
43	Guided tutorial	As an Icelandic traveler, I want a guided tutorial of the summary screen so that I can learn it quickly.	L	Tutorial auto appear on first use. Each step has a skip option.	C
44	Dark mode	As an Icelandic traveler, I want dark mode support so that it's comfortable to read at night.	L	User can change to dark mode in system settings. All screens render in dark mode.	C

ID	Name	Description	Story Points	Acceptance Criteria	Priority
45	Country auto detect	As an Icelandic traveler, I want country auto-detect to preselect language so the I don't have to switch manually.	XL	User can share location. Summary changes language according to the country the user is in.	C

Appendix D: Risk Analysis

D.1 Risks

Risk	Scope Triangle	Event #	Event	Probability	Impact	Score	Mitigate, Why or How
<i>Technical</i>							
Tech	Quality	TQ01	Insufficient translation tool	M	H	12	M, mitigate later if errors occur
Tech	Quality	TQ02	Security risk (data breach)	M	H	12	Y, Sensitive health data compliance requires mitigation
Tech	Scope	TS01	Missing API functionality	H	VH	20	Y, Critical, dependency, need fallback plans
Tech	Time	TT01	Complicated API	M	M	9	M, planning and monitor progress
Tech	Res	TR01	Learning curve/integrating for new technologies	M	L	6	M, Training, adapt new skill
Tech	Res	TR02	Hardware failure/system downtime	L	M	6	N,rely on backups and standard IT support if it occurs
<i>Financial</i>							
Fin	Cost	FC01	Budget overruns	M	H	12	M, Establish cost buffer and review expenses each sprint to stay within budget
Fin	Cost	FC02	External contractor cost increases	M	M	6	M, lock in cost negotiations
<i>Organizational</i>							
Org	Quality	OC01	Legal risk, violations of regulations	VL	VH	5	Yes, to prevent risk of penalties
Org	Quality	OC02	Communication difficulties	M	M	9	Monitor, check-ins
Org	Res	OR01	Shortage of manpower	M	H	12	Y, Harms timeline, mitigate by cross-training

Table D.1: Risk Events and associated Probability, Impact, Risk score and Mitigation strategy. Risk categories: Technical, Organizational, Financial, Social, Project Management, External. Scope triangle dimensions: Scope, Time, Quality, Cost, Resources. Probability/Impact legend: Very High (VH), High (H), Medium (M), Low (L), Very Low (VL). Mitigation legend: Yes (Y), Monitor (M), No (N).

Risk	Scope Triangle	Event #	Event	Probability	Impact	Score	Mitigate
Org	Res	OR02	Change of leadership	L	M	6	M, Maintain detailed documentation to ensure smooth handover and project continuity.
Org	Res	OR03	Inadequately skilled members	L	M	6	M, require training during onboarding
Org	Res	OR04	Lack of resource support	L	H	8	M, allocate/identify recourse and logistics from start
<i>Social</i>							
Soc	Res	SR01	Illnesses	M	M	9	M, Provide flexible scheduling
Soc	Res	SR02	Family crisis	L	M	6	M, flexible scheduling, cross-training
Soc	Res	SR03	Daily mood of team members	M	L	6	N, minor day-to-day variation, not significant enough to require formal mitigation
<i>Project management</i>							
Proj. Man	Time	PT01	Schedule conflicts/ project deferral	H	M	12	Y, proactive project management
Proj. Man	Time	PT02	Aggressive schedule, overlapping deadlines	M	M	9	M, Agile planning
Proj. Man	Scope	PS01	Scope creep, additional features risk derailing focus	H	VH	20	Y, maintain clear and strict scope control
Proj. Man	Scope	PS02	Unclear requirements, unclear job description	M	H	12	Y, Provide clear alignment
Proj. Man	Res	PR01	Key role member quitting or change of career	M	VH	15	Y, documentation & cross-training
<i>External</i>							
Ext	Scope	ES01	Regulation mismatch, summaries not accepted abroad	M	H	12	Y, engage regulators early, test trials
Ext	Scope	ES02	Change in regulations, standard	VL	H	4	M, assess legislative changes
Ext	Res	ER01	Public health crisis, pandemic, disrupting work	L	VH	10	M, remote collaboration tool
Ext	Time	ET01	Travel constrictions (weather)	M	L	6	N, remote collaboration tool

D.2 Risk Matrix

	Very Low (1)	Low (2)	Medium (3)	High (4)	Very High (5)
Very High (5)	5	10	15	20	25
High (4)	4	8	12	16	20
Medium (3)	3	6	9	12	15
Low (2)	2	4	6	8	10
Very Low (1)	1	2	3	4	5

	Ignore
	Consider
	Take action

Table D.3: Matrix for calculation Risk score ranging from 1–25 from Probability and Impact